$R = 8.13\,J/(mol\cdot K)$

Set 10 - 热学2

作业10A 7.23 7.31

作业10B 8.5 8.7 8.8 8.13 8.15

7.23 $P = 1.33\times10^{-3}P_0$ $T = 300K$ $d = 3\times10^{-10}m$

$PV = \nu RT$ $n = \frac{\nu N_A}{V} = \frac{PN_A}{RT} = 3.38\times10^{17}$

$\lambda = \frac{1}{\sqrt{2}n\pi d^2} = 7.40m$ >> 真空管线度 实际中的自由程为真空管线度

7.31 (1) $V = Nb = 7.8\times10^{-4}m^3$

(2) $\frac{M}{\mu}N_A l^3 = V \Rightarrow l = 5.0\times10^{-10}m$

(3) $p_i = a(\frac{N}{V})^2 = 9.3\times10^7 P_0$

8.5 (1) $Q_1 = \Delta E + W_1 \Rightarrow \Delta E = 224J$

$Q_2 = \Delta E + W_2 \Rightarrow Q_2 = 266J$

(2) $-\Delta E = Q_3 + A_1 \Rightarrow Q_3 = -308J$

系统放热308J

8.7 $\Delta E = \nu\frac{i}{2}R\Delta T \Rightarrow \Delta E = 124.95J$

$\Delta E = Q + A \Rightarrow Q = -87.05J$

$C = \frac{dQ}{\nu dT} \Rightarrow C = -8.705J/(mol\cdot K)$

8.8 $P_1 = 1.013\times10^5 P_a$ $V_1 = 8.2L$ $T_1 = 300K$ $i = 5$ $T_2 = 400K$

(1) $PV = \nu RT \Rightarrow \nu = 0.34mol$

$\Delta E = \nu\frac{i}{2}R\Delta T \Rightarrow \Delta E = 692.22J$

$Q = \Delta E \Rightarrow Q = 692.22J$

(2) $P_1V_2 = \nu RT_2$

$W = P_1(V_2 - V_1) = \nu R(T_2 - T_1) \Rightarrow W = 270.42J$

$Q = \Delta E + W \Rightarrow Q = 968.64J$

压强不变的热更多，因为需对外做功

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8.13 $m=1kg$ $P_1=1\times10^5Pa$ $T_1=293K$ $P_2=1\times10^6Pa$ $\rho=1.19kg/m^3$

$V_1=\frac{m}{\rho}$

$P_1V_1=P_2V_2=\nu RT$ $\Rightarrow A=-\nu RT\ln\frac{V_2}{V_1}$

$A=-\int p\,dV$ $=-P_1\frac{m}{\rho}\ln\frac{P_1}{P_2}$

$=1.93\times10^5J$

8.17 $V_1=10L$ $T_1=273K$ $P_1=1.013\times10^6Pa$ $P_2=1.013\times10^5Pa$

(1) $\gamma=1+\frac{2}{i}=\frac{7}{5}$

$P_1V_1^{\frac{7}{5}}=P_2V_2^{\frac{7}{5}}$ $\Rightarrow \begin{cases} V_2=51.8L \\ T_2=141.4K \end{cases}$

$P_2V_2=\nu RT_2$

$P_1V_1=\nu RT_1$

$Q=\Delta E+W=0$ $\Rightarrow W=1.22\times10^4J$

$\Delta E=\nu\frac{i}{2}R\Delta T=\frac{i}{2}(P_2V_2-P_1V_1)$

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